

Aman Kumar

IIT Roorkee Graduate '26

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Professional Summary

IIT Roorkee graduate (B.Tech + M.Tech, Geophysical Technology, 2026) with hands-on experience building Generative AI systems, RAG pipelines, and agentic workflows using Python, LangChain, smolagents, and HuggingFace. Interned at an AI startup contributing to backend APIs and LLM-powered ticket automation. Strong competitive programmer with a Codeforces rating of 1550 (Specialist), CodeChef rating of 2100 (5-Star), and 1500+ problems solved across platforms. Seeking a software engineering or AI/ML role to build intelligent, production-ready systems.

Education

Year	Degree / Examination	Institution / Board	CGPA / %
2026	B.Tech + M.Tech — Geophysical Technology	Indian Institute of Technology, Roorkee	6.5
2021	Intermediate (Class XII)	Gaya College (BSEB), Gaya, Bihar	77.40%
2019	Matriculate (Class X)	B.L Indo Anglian Public School (CBSE)	86.80%

Technical Skills

Languages	Python, C++, JavaScript, Swift
AI / ML Frameworks	LangChain, LangGraph, smolagents, LightRAG, HuggingFace Transformers, FAISS, ChromaDB
Databases & Graphs	MySQL, Neo4j, Firebase Firestore
Developer Tools	Git, GitHub, VS Code, Xcode, Flask, Gradio, Streamlit
Core Concepts	Large Language Models (LLMs), RAG, Agentic AI, Knowledge Graphs, Vector Embeddings, REST APIs
Coursework	Data Structures & Algorithms, Computer Networking, Operating Systems, Generative AI (LangChain & HuggingFace), HuggingFace AI Agents

Work Experience

Software Development Intern | CallSure AI

Jun 2025 – Jul 2025

- Developed a RAG-based query resolution pipeline using LangChain and FAISS to retrieve relevant historical support resolutions and documentation for incoming customer queries, enabling context-aware automated responses.
- Implemented a semantic query classification workflow using HuggingFace embeddings to automatically score and categorize incoming requests by urgency, replacing a manual rule-based priority system.
- Built a priority routing mechanism where high-severity queries triggered an internal Flask API for immediate escalation to support teams.
- Designed the Flask API to store and manage structured ticket data in MySQL, enabling efficient retrieval of customer history and resolution records during query processing.

Projects

LightRAG Document Intelligence System | Python, LightRAG, Neo4j, FAISS, Streamlit

Jan 2025 – Mar 2025

- Architected a document Q&A system via LightRAG that constructs a dual-level knowledge graph (entity + relationship) over corpora, enabling pinpoint fact lookup and broader theme-level queries unsupported by standard RAG.
- Cut LLM token consumption by ~65% vs. Microsoft GraphRAG by leveraging LightRAG's dual-level indexing; backed retrieval with a Neo4j graph store and FAISS vector index for multi-hop reasoning.
- Shipped a Streamlit interface supporting PDF upload, natural language querying, and full source-citation tracing; published the codebase to GitHub with detailed setup documentation.

Autonomous Code Review & Debug Agent | smolagents, LangGraph, FAISS, Gradio

Feb 2025 – Apr 2025

- Engineered a CodeAct-style autonomous agent with HuggingFace smolagents and LangGraph that analyses a code snippet or small repository, flagging bugs, anti-patterns, and security vulnerabilities across 3 severity tiers.
- Equipped the system with 3 custom tools — a code-execution sandbox, a RAG-powered documentation retriever (FAISS + HuggingFace embeddings), and a web-search tool — enabling self-guided debugging without human intervention.
- Deployed a Gradio interface where users submit code and receive an annotated report with severity labels and fix suggestions; the pipeline operates via a multi-step ReAct reasoning loop.

LostFound — Campus Lost & Found Board | Python, Flask, SQLite, HTML/CSS

Oct 2024 – Nov 2024

- Developed a full-stack web application using Flask and SQLite to digitise the lost-and-found process for 5000+ students at IIT Roorkee, replacing informal WhatsApp groups with a structured, searchable platform.
- Engineered RESTful route handlers in Flask for item submission, category-based filtering, and a claim/resolve workflow; designed a normalised SQLite schema with Flask-SQLAlchemy ORM for efficient data retrieval.
- Built a responsive, mobile-friendly frontend using Jinja2 templates and HTML/CSS; deployed on the campus local network enabling real peer-to-peer usage and published to GitHub with full setup documentation.

Achievements

- Attained Codeforces rating of 1550 (Specialist) and CodeChef rating of 2100 (5-Star); solved 1500+ algorithmic problems across platforms with consistent participation in weekly rated contests.
- Competed in institute-level AI/ML hackathons at IIT Roorkee, delivering working LangChain and Flask prototypes within tight deadlines.
- Secured top 1% rank in both JEE Mains and JEE Advanced among 1.2M+ candidates; earned admission to IIT Roorkee — one of India's premier engineering institutes; scored 86.80% in Class X and 77.40% in Class XII.

Hobbies & Interests

Football & Cricket: Enjoy playing and following both sports; regular viewer of international tournaments. **Travelling:** Like exploring new places and experiencing different cultures. **Gaming:** Regular PUBG player — enjoy team-based strategy and quick decision-making under pressure. **Tech Trends:** Follow Generative AI, agentic AI, and open-source LLM research through arXiv, newsletters, and podcasts.